

COURSE CODE	COURSE TITLE	L	T	P	TW & SL	TH	C
UEC3942	INTRODUCTION TO SENSORS AND ACTUATORS (Other than ECE Department)	45	0	0	45	90	3

PREAMBLE

This subject deals with the study and understanding of different types of sensors and actuators. It defines the characteristics of micro sensors and micro actuators and gives an in-depth knowledge about the operation of various sensors and its applications in the real time scenarios. The subject also deals with the characterization of MEMS/NEMS devices and its available packaging techniques.

OBJECTIVES

- To study different types of sensors and actuators
- To gain knowledge on the type of optical sensors to be used for practical applications
- To understand the design concepts of micro sensors and micro actuators
- To understand the packaging and characterization of MEMS/NEMS

UNIT I	STRAIN, PRESSURE AND TEMPERATURE	9
Introduction, Stress & Pressure sensors: Resistance strain gauge, piezoelectric strain gauge, characteristics - Fibre-optic sensor - Pressure gauges - Temperature Sensors: Bimetallic strip, thermocouples, Resistance thermometers, thermistors, bolometer, Pyroelectric detector.		

UNIT II	OPTICAL SENSORS	9
Colour temperature, light flux, photo sensors, photomultiplier, photoresistor and photoconductors, photodiodes, phototransistors, photovoltaic devices, fibre optic sensors, electro optic sensors, fibre-optic applications, light transducer, solid-state transducers and liquid crystal devices.		

UNIT III	MICRO SENSORS	9
Acoustic sensor - Quartz crystal microbalance, surface acoustic wave, Flexural plate wave, shear horizontal; Vibratory gyroscope, Capacitive and Piezo Resistive Pressure sensors.		

UNIT IV	MICRO ACTUATORS	9
Electrostatic actuators - parallel plate capacitor, Interdigitated finger capacitor, piezoelectric actuators, Thermal actuators, Actuators using shape memory alloys - Microgrippers, Micromotors, Microvalves, Micropumps		

UNIT V	PACKAGING AND CHARACTERIZATION	9
Micro / nano systems packaging, Essential packaging technologies, Selection of packaging materials; SEM, TEM, AFM, STM, Spectroscopic techniques for Nano characterization.		

TOTAL PERIODS	45
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COURSE OUTCOMES

On successful completion of this course, students will be able to
CO1 Select sensors for the measurement of physical variables for real time applications
CO2 Use sensors for optoelectronic applications
CO3 Elucidate and apply sensors for engineering applications
CO4 Elucidate and apply actuators for engineering applications
CO5 Examine the packaging and characterization requirements

TEXT BOOKS

1. Ian R Sinclair, Sensors and Transducers, Newnes publishers, Third Edition, 2001. (Unit I & II).
2. Tai-Ran Hsu, MEMS and Microsystems Design and Manufacture, Tata McGraw Hill, 2017. (Unit III to V).

REFERENCE BOOKS

1. Doebelin E O, Measurement Systems, Application and Design, McGraw Hill, Fifth Edition, 2004.
2. Murty B.S, Shankar P, Raj B, Rath, B.B, Murday J, Textbook of Nanoscience and Nanotechnology, Springer publishing, 2013
3. Vinod Kumar Khanna, Nanosensors: Physical, Chemical, and Biological, CRC press, First Edition, 2012.
4. Chang Liu, Foundations of MEMS, Pearson education India limited, Second Edition, 2006.
5. Ramon Pallás Areny and John G. Webster, Sensors and Signal conditioning, John Wiley and Sons, Second Edition, 2000.

CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	2	1	1	1			3	2	1
CO2	3	2	3	2	3	2		1				3	3	2
CO3	3	3	3	2	2	2	1	1	1		1	3	2	2
CO4	3	3	3	2	3	2	1	1			1	3	2	2
CO5	2	2	2	3	2	2	2				1	2	1	3

